

Lunar Surface Analysis via XRF

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1 Introduction

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- **Goal:** Identification and quantification of chemical elements on the lunar surface to support future exploration.
- **The Challenge:** Remote sensing via X-ray Fluorescence (XRF) produces complex, noisy, and high-dimensional spectral data.
- **Methodology:** Using Machine Learning to automate and refine the interpretation of these spectra.
- **Data Strategy:** Development of a custom dataset and utilization of multiel-spectra for high-fidelity synthetic data generation.

XRF Spectroscopy Fundamentals

- High-energy X-rays/Gamma rays irradiate the sample.
- Secondary X-rays (fluorescence) are emitted, unique to each element.
- The detector captures a spectrum where energy peaks correspond to specific atoms.

